HOMEWORK EXERCISE 2 SOLUTIONS

1) THE SAMPLE MEAN IS THE MEAN OF A SUBSET (NOT ALL) OF OBSERVATIONS FROM THE POPULATION; THE POPULATION MEAN IS THE MEAN OF ALL OBJECTS OF THE POPULATION.

SAMPLE MEAN ($\bar{x}$) CAN BE COMPUTED; POPULATION MEAN (μ) CANNOT SINCE IT IS IMPOSSIBLE TO OBTAIN ALL OBJECTS TO BE MEASURED.

2) a) 503.4 IS A STATISTIC

WE DENOTE IT $\bar{x} = 503.4$

b) GIVEN WE RECORDED THE STANDARD DEVIATION OF THE SAMPLE OF BIRDS, WE HAVE S, NOT σ, SO USE THE t - CONFIDENCE INTERVAL FORMULA.

98% CI $\Rightarrow \alpha = 0.02$ $\quad n = 25$ $\quad \bar{x} = 503.4$

$t_{\alpha/2} = t_{0.01, 24}$ $\quad df = 24$ $\quad S = 15$

$= 2.492$

98% CI: $503.4 - 2.492 \frac{15}{\sqrt{25}}$, $503.4 + 2.492 \frac{15}{\sqrt{25}}$

$503.4 - 7.476$, $503.4 + 7.476$

495.924 , 510.876

WE ARE 98% CONFIDENT THAT THE TRUE MEAN WEIGHT OF BIRDS IN THIS POPULATION IS BETWEEN 495.924 g AND 510.876 g.

2 c) NO, SINCE THE 98% CONFIDENCE INTERVAL DOES NOT CONTAIN THE VALUE 517.0 g, IT WOULD NOT BE REASONABLE TO CONCLUDE THE TRUE MEAN WEIGHT OF THESE BIRDS IS 517.0 g.